higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL WORKMANSHIP
NQF LEVEL 4

8 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 Electrocution is death✓ caused by an electric shock.✓ (2)

1.2 1.2.1 100 kΩ
1.2.2 1 kΩ
(2 × 1) (2)

1.3 1.3.1
- touch/step potential
- hand/foot path (Any 1 × 2) (2)

1.3.2 In a touch/step potential contact, current travels from one hand✓ through the heart,✓ down the leg,✓ and out through the foot.✓ (4)

1.4
- Use the Work Permit or Permit to Work system
- Use the lockout and tag-out procedure (Any 1 × 2) (2)

1.5
- Probability
- Severity
- Exposure (3)

1.6 1.6.1 D
1.6.2 C
1.6.3 A
(3 × 1) (3)

1.7
- A place for everything✓ and everything in its place.✓
- Keeping things in their proper place✓, no unnecessary items lying around.✓ (Any 1 × 2) (2)

[20]

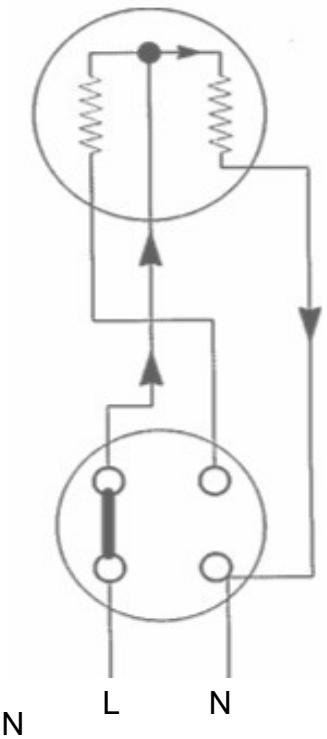
QUESTION 2

2.1 2.1.1
- Infrared sensors
- Passive infrared sensors (2)

2.1.2 Infrared sensors detect the motion of heat from one area to another.✓ As they operate on a line-of-sight principle, they do not work well over long distances or in rooms with obstructions.✓

Passive infrared sensors emit sound waves and are very sensitive to motion, working well when there are obstructions.✓ Motion of any kind will disrupt the reflected frequency, which detects movement.✓ (2 × 2) (4)

2.2



One mark for supply L and N
One mark for indicating link in switch linking live to circuit
One mark for indicating connection in plate
One mark indicating one element used
One mark for showing correct current flow (5)

Medium Heat Stove Switch

2.3 When the temperature of the water rises, the brass tube expands more than the invar rod,✓ which will cause the contact to open,✓ thereby opening the circuit and hence preventing any current flow.✓ (3)

2.4 2.4.1 distribution board (2 × 1) (2)

2.4.2 neutral (1)

2.5
- Phase-to-earth
- Phase-to-neutral
- Phase-to-phase
- All three-phases short-circuit
- Phase-to-phase-earth short-circuit
- All three-phases short-circuit to earth (Any 3 × 1) (3)

[20]

QUESTION 3

3.1 A light meter (1)

3.2
- By using a large number of lamps
- By positioning lamps correctly (2)

3.3 3.3.1 Industrial uses

3.3.2 Drawing office or Workshops

3.3.3 Rooms with high ceilings

3.3.4 Interior decorating

(4 × 1) (4)

3.4 3.4.1 Incandescent lamps (1)

3.4.2
- Normal incandescent or tungsten lamps
- Tungsten halogen or halogen lamps (2)

3.4.3 Approximately 1 000 hours (1)

3.5
- Traffic signals
- Vehicle lights
- Aviation lighting
- Domestic lighting
- Optical encoders
- Alphanumeric displays
- Remote controls
- Level indicators
- High-voltage indicators
- Security systems (Any 2 × 1) (2)

3.6 When discharge lighting✓ is used near rotating machinery,✓ there is risk that the rotating parts may appear to be stationary.✓ (3)

3.7 3.7.1
- Glow starters
- Thermal starters (2)

3.7.2
- It completes the circuit initially for pre-heating the electrode.
- It opens the circuit to provide a voltage kick for ignition. (2)

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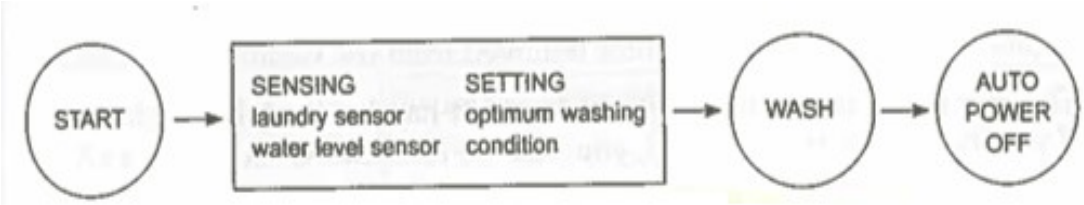
QUESTION 4

4.1 4.1.1 It has basic insulation throughout✓ and is provided with an earthing terminal or earthing contact.✓ (2)

4.1.2 It has double insulation or reinforced insulation✓ or both throughout.✓ (2)

4.2
- Know what the appliance is used for
- Know where the appliance is used
- Know how the appliance works (3)

4.3
- The cable length
- The type of cable
- The cable current rating
- The cable insulation voltage
- The cable temperature rating
- The cable rigidity or flexibility
- Security systems (Any 2 × 1) (2)



4.4
- Ensure that the timer control is in the off position.
- Bridge the fuse to the thermostat supply wire. If the oven still does not work, then the timer is not faulty.
- If the timer is faulty, replace it with a new unit. (3)

4.5
- At the customer's premises
- In your own workshop
- In the industrial environment or factory (3)

4.6 ✓ ✓ ✓ ✓ ✓

Fuzzy logic control programme (5)
[20]

QUESTION 5

5.1 Planning a task refers to an intended method of getting from one set of tasks✓ to another in a systematic and logical way.✓ (Any relevant answer) (2)

5.2
- Insulation resistance tester (Megger)
- Ohmmeter
- Clamp-on meter (Tong tester)
- Voltmeter
- Voltage tester
- Multimeter (Any 5 × 1) (5)

5.3 5.3.1
- Isolate the motor according to safe working practices.
- Use a soft brush and clean cotton rags to remove excess dust from the surface of the motor.
- Remove the fan cover/cowling and clean the cover using compressed air
- Clean the fan and cooling fins using compressed air.
- Inspect the terminal box and remove any dust and moisture that might have accumulated in the terminal box.
- Check if the terminals are free of rust and in good condition. (6)

5.3.2
- Short-circuit and open-circuit test (continuity test)
- Insulation resistance test between windings
- Insulation resistance test between each winding and earth (3)

5.4
- Uncertainty about the load size
- Allowance for further load growth
- Rounding up to the next size motor
- Availability (4)

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TOTAL: 100

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